Cloud Forensics Challenges in Bangladesh: A PRISMA Systematic Review with AI-Enhanced Resolution Strategies

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***Abstract*—Bangladesh's accelerating cloud adoption—over 100 million internet-enabled devices and more than $1.2 billion in public-sector cloud procurement since 2020—has outpaced any corresponding update to the legal and procedural infrastructure governing digital investigations. The country's investigative statutes (ICT Act 2006, Digital Security Act 2018, Cyber Security Act 2023) were drafted for physical endpoint devices and are silent on cloud-hosted evidence, statutory log preservation, and cross-border jurisdictional complexity. The result is a growing and consequential gap: the Cyber Tribunal records a disproportionately high acquittal rate on cloud-sourced evidence cases, driven not by disputed facts but by chain-of-custody failures and Section 65B admissibility deficiencies.**

This paper addresses that gap through a PRISMA 2020-compliant review of 16 open-access publications (2021–2026), identifying five challenge clusters (C1–C5) and mapping each to a validated AI resolution strategy with quantitative evidence. The resulting *AI-Enhanced Cloud Forensic Readiness Framework for Bangladesh* (AI-CFRF-BD) provides a phased, locally implementable pathway for Bangladesh's law enforcement and policymakers.s.

***Index Terms*—cloud forensics, digital evidence, Bangladesh cyber law, PRISMA systematic review, AI forensics, blockchain chain-of-custody.**

I. Introduction

Bangladesh has undergone a rapid digital transformation driven largely by mobile connectivity. By 2023, over 66.94 million citizens used the internet regularly, with more than 100 million internet-enabled devices in circulation [15], and public-sector cloud procurement surpassing $1.2 billion between 2020 and 2024 [16]. Government ministries, financial institutions, and private enterprises have migrated critical workloads to cloud platforms at pace, yet the country's investigative framework was designed for seized hard drives and physical devices—not for evidence residing on foreign cloud servers [12]. Cloud forensics demands fundamentally different assumptions about data custody, access rights, evidence provenance, and jurisdictional authority [13]. Bangladesh's Cyber Security Act 2023 (CSA 2023), despite establishing a dedicated Cyber Tribunal, is entirely silent on cloud-hosted evidence, statutory log preservation, and cross-border jurisdictional complexity—leaving a growing and consequential enforcement gap.

A. Motivation

The practical cost of this gap is visible in the Cyber Tribunal's case record. Cloud activity logs are automatically deleted on 30-to-90-day retention cycles, yet Bangladesh has no statutory data-hold mechanism to compel CSPs to suspend deletion while a court order is obtained [10]. Section 65B of the Evidence Act—the sole mechanism for certifying digital evidence admissibility—was drafted for records derived from devices physically in the investigator's possession, a precondition almost never met for remotely hosted cloud logs [10], [11]. Uzzal's review documents a consistent pattern: acquittals driven not by disputed facts but by broken custody chains, malformed certificates, and forensic reports that ignore the cloud storage layer [11]..

B. Research Gap

No prior study applies PRISMA 2020 to systematically map cloud forensics challenges within Bangladesh's legal context. Alshabibi et al.'s global SLR (31 studies) includes no Bangladesh-specific analysis, omits the Section 65B barrier, and predates the 2025 legislative instruments [1]. Bangladesh-authored works [10], [13], [15] address fragments without mapping challenges to validated AI tools. No prior work simultaneously addresses all five challenge dimensions with Bangladesh relevance; the present review is the first to do so.

C. Research Questions

Four RQs were pre-registered before any database search commenced:

RQ	Question
RQ1	What technical challenges do cloud environments pose to digital forensics in BD?
RQ2	What legal barriers prevent admissibility of cloud-sourced evidence under Bangladesh's legal framework?
RQ3	How do AI-enhanced tools address the challenges in RQ1 and RQ2?
RQ4	What implementable framework can improve cloud forensic readiness for BD law enforcement?

D. Contributions

- The first PRISMA 2020 SLR mapping cloud forensics challenges to Bangladesh's legal and institutional context

(16 open-access studies, 2021–2026); comparative analysis of prior work in Table I.
- A five-cluster challenge taxonomy (C1–C5) cross-referenced against concrete gaps in the CSA 2023, Evidence Act Section 65B, MLAT mechanism, and CSP data-sharing frameworks (Table II).
- The AI-CFRF-BD framework: five validated AI tools mapped to the five clusters, each deployable within Bangladesh's existing legal authority (Figs. 1–2).

II. Literature Review

A. Cloud Forensics: Global Landscape

Cloud computing's defining features—on-demand provisioning, elastic scaling, resource pooling across IaaS, PaaS, and SaaS layers, support for public, private, and hybrid deployments—are precisely what make it difficult to investigate [13]. The business advantages and the forensic disadvantages come from the same source.

Alshabibi, Liatsis, and Emhemed's 2024 systematic review of 31 studies is the most comprehensive treatment of this problem to date [1]. They identify seven technical barriers—volatile data, multi-tenancy, encryption, virtualisation, jurisdictional fragmentation, insufficient readiness, inadequate tooling—and five legal ones: inadequate legislation, admissibility uncertainty, jurisdiction conflicts, no international framework, and privacy constraints. Worth noting is that these categories are not independent. A single investigation can hit several simultaneously, and the legal barriers often make the technical ones worse.

Two theoretical contributions help frame what follows. Karagiannis and Vergidis coined the "Power of Disposal" concept: the party with legal authority to grant access to cloud data is the service provider, not the state issuing the warrant [8]. That inversion of the traditional evidence-access model has no clear analogue in most domestic legal frameworks, including Bangladesh's. Separately, Prakash et al. mapped a five-stage forensic lifecycle and found that evidence preservation—not analysis, not presentation—is where cloud investigations most often fail [3]. The window between detecting an incident and losing the logs to routine deletion is, in many cases, shorter than the time needed to obtain a court order.

B. Bangladesh Legal and Institutional Context

Bangladesh's digital investigation framework rests on three statutes: the ICT Act 2006, the Digital Security Act 2018, and the CSA 2023. All three were drafted for endpoint devices and are silent on cloud-hosted evidence [10]. Section 65B of the Evidence Act provides the only statutory mechanism for digital evidence admissibility, but its custody certification procedure presupposes that the evidence originated from a device in the investigator's direct possession—a precondition almost never met for cloud logs [15]. Uzzal's review of Cyber Tribunal judgments identifies three recurring admissibility failures: broken chains of custody, missing or malformed Section 65B certificates, and forensic reports that do not account for the cloud storage layer between the alleged offence and the physical exhibit presented in court [11]. Ahmed and Arifuzzaman confirm that CSA 2023, despite creating the Cyber Tribunal and updating cybercrime definitions, carries these structural gaps forward unchanged [12]. Subsequent 2025 legislation has compounded rather than resolved the problem: the Personal Data Protection Ordinance 2025 imposes data localisation duties on cloud operators but provides no forensic access mechanism; the Cyber Security Ordinance 2025 expands Tribunal jurisdiction without adding a preservation power; and the National Digital Governance Framework 2025 mandates cloud-first adoption across the public sector with no accompanying forensic readiness requirement. Sharmin and Rahman's survey of 52 Bangladesh Police officers documents six recurring operational limitations: evidence accuracy shortfalls, collection process failures, low conviction rates on digital evidence, geolocation difficulties, reactive rather than preventive posture, and limited technical capacity for novel investigation types [15].

C. Cross-Border Evidence Access

When cloud data is stored on foreign infrastructure, it falls under that country's laws. Accessing it for a domestic investigation nominally requires an MLAT request—a mechanism designed for physical evidence transfer between cooperating states. Akilal and Kechadi's analysis of 47 jurisdictions found average MLAT processing times of six weeks to ten months, strongly correlated with bilateral relationship closeness and the presence of a specialist digital evidence track [9]. Neither condition currently applies to Bangladesh. Ekwempu's 2026 analysis of *United States v. Microsoft Corp.* and the Yahoo Data Breach further demonstrates that even court-ordered access to extraterritorial cloud data is contested, and that resolution required years of primary legislation [14]. Rana's comparative study of Colombia, India, and the United States shows that domestic legislative adaptation to the cloud paradigm—not the extension of endpoint-focused provisions—is the only effective remedy [16].

D. Research Gap and Motivation

Table I positions the eight most relevant prior studies against the five challenge clusters identified in this review. Three things stand out.

Akter and Rahman [13] come closer than anyone to covering the full picture—they address C1 through C4—but their work predates the CSA 2023 and proposes no AI-based resolution strategy. The four AI-tool papers [5]–[7], [9] are technically strong, but every one of them was validated in US or EU environments. None grapples with Section 65B. None considers what it means to operate in a jurisdiction with no statutory data-hold mechanism. The Bangladesh-specific studies [10], [11], [15] document the failures well but stop at diagnosis. They do not connect the observed problems to the AI-based tools available in the international literature.

What is missing, in short, is a study that takes the Bangladesh legal context seriously, applies reproducible methodology, covers all five challenge dimensions, and arrives at actionable recommendations grounded in peer-reviewed evidence. The present review addresses it by applying

TABLE I
COMPARATIVE ANALYSIS OF MOST RELEVANT PRIOR STUDIES AGAINST THE FIVE CHALLENGE DIMENSIONS

Ref	Focus & Contribution	C1	C2	C3	C4	C5	AI Tech.	BD Rel.	Method	Year
[1]	Alshabibi et al. — SLR, 7 tech. + 5 legal barriers	✓	✓	■	■	✓	ML survey	×	SLR	2024
[5]	Alharthi & Garcia — CIAF LLM (93% F1)	✓	✓	■	×	■	LLM/GenAI	■	Framework	2025
[6]	Igonor et al. — blockchain CoC, 14/17 jurisd.	×	■	✓	✓	×	Blockchain	■	Prototype	2025
[7]	Tageldin & Venter — ML-CF, ISO/IEC 27043	✓	✓	■	×	✓	ML/LSTM	■	SLR	2023
[9]	Akilal & Kechadi — CLERMS, 40–70% MLAT	✓	×	×	✓	■	NLP/Log	■	Prototype	2025
[12]	Ahmed & Arifuzzaman — CSA 2023 BD gap	✓	✓	✓	■	✓	Statistical	✓	Survey	2025
[13]	Akter & Rahman — cloud forensic dims., UIU	✓	✓	✓	✓	■	Process ML	✓	SLR+Model	2023
[15]	Sharmin & Rahman — BD Police survey (52)	■	×	✓	✓	✓	Statistical	✓	Survey	2025
[P]	**This study — AI-CFRF-BD (PRISMA SLR)**	✓	✓	✓	✓	✓	LLM+ML+BC	✓	PRISMA	**2026**

✓ = Fully addressed ■ = Partially × = Not addressed BD Rel. = Bangladesh relevance [P] = This study (framework proposal level)

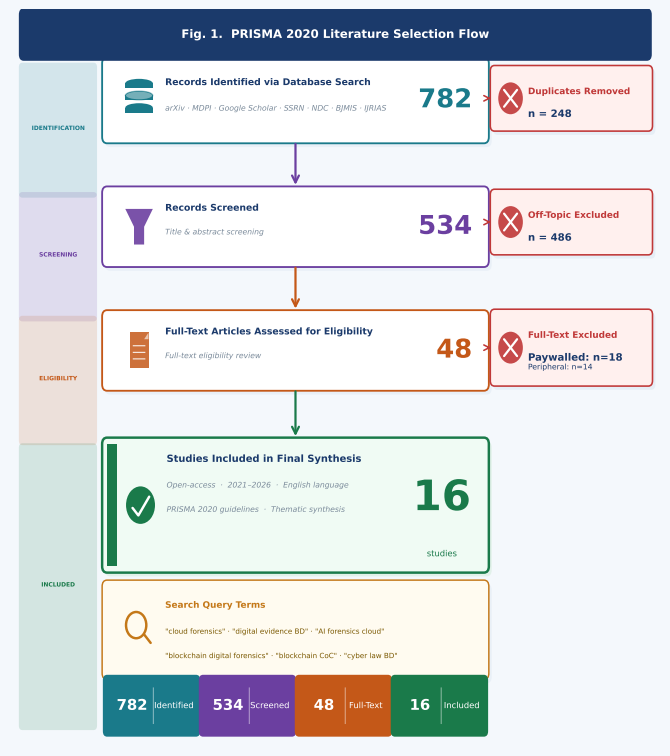


Fig. 1. PRISMA 2020 literature selection flow: 782 records identified; 16 studies included after duplicate removal, screening, and eligibility assessment.

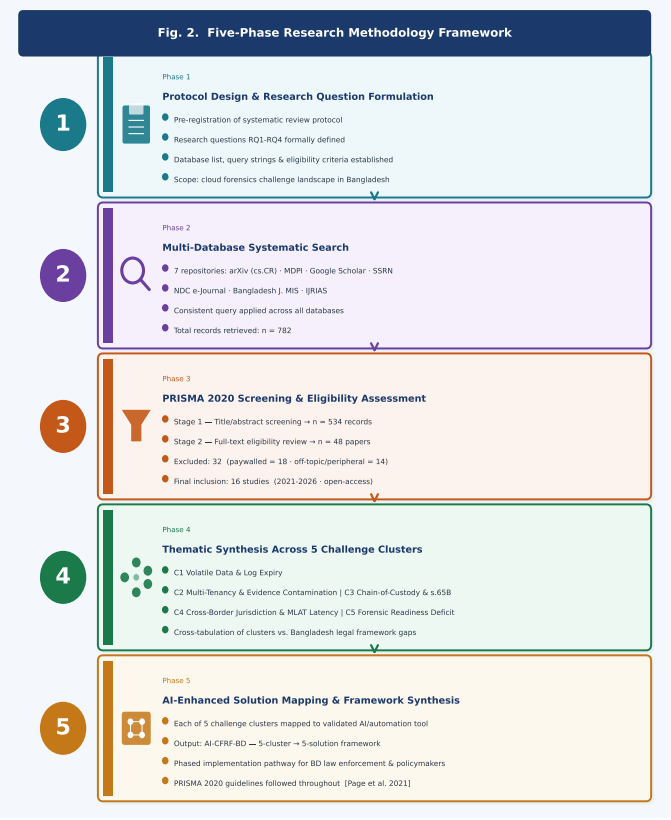


Fig. 2. Five-phase research methodology: protocol design, multi-database search, PRISMA screening, thematic synthesis, and AI-enhanced solution mapping.

PRISMA 2020 methodology to produce the first such synthesis.

III. METHODOLOGY

This review adopts a PRISMA 2020-guided systematic literature review (SLR) design [18], selected because it provides explicit, auditable procedures for every stage of source selection and a transparent structure that distinguishes this study from narrative reviews and enables future replication and updating as the legislative landscape evolves. The PRISMA 2020 selection flow is shown in Fig. 1 and the complete five-phase research process is illustrated in Fig. 2.

A. Search Strategy

Seven repositories were searched: arXiv (cs.CR) for technically current AI forensics preprints—CIAF and CLERMS would have been missed without it—MDPI Open Access for peer-reviewed forensics journals, Google Scholar and SSRN for broader coverage, and the NDC e-Journal, BJMIS, and IJRIAS for Bangladesh-specific policy literature that rarely surfaces in international databases. Six query strings were applied uniformly: *"cloud forensics"*, *"digital evidence Bangladesh"*, *"AI forensics cloud"*, *"blockchain digital forensics"*, *"blockchain chain-of-custody"*, and *"cyber law BD"*. The 2021–2026 window was chosen to align with the post-DSA 2018 period and to capture studies engaging with the CSA 2023 environment. The search yielded 782 records.

TABLE II
TAXONOMY OF CLOUD FORENSICS CHALLENGES, AI SOLUTIONS, AND BANGLADESH CONTEXT

ID	Challenge & Root Cause	BD Legal Gap	AI Resolution Tool	BD Impact	Metric
C1	**Volatile Data & Log Expiry.** CSP logs auto-deleted (30–90 days); detection window closes before court order issues.	No data-hold statute; ICT Act 2006 & CSA 2023 silent.	**CIAF** (Alharthi & Garcia, 2025): LLM hash-stamps logs before deletion; no CSP cooperation required.	Evidence stamped in window; no legislative change; deploys with NTMC.	93% F1
C2	**Multi-Tenancy & Evidence Contamination.** Shared infra; container isolation weak; one tenant extraction risks co-resident exposure.	BD criminal code silent on commingled evidence; CSA 2023 does not address this.	**ML-CF** (Tageldin & Venter, 2023): RF, LSTM, CNN isolate forensic artefacts; ISO/IEC 27043-aligned.	Artefacts from NTMC feeds; no new legislation; ISO/IEC 27043 aligned.	ISO/IEC 27043
C3	**Chain-of-Custody & s.65B Admissibility.** No CSP-to-court custody record; s.65B drafted for physical devices.	s.65B precondition unmet for cloud logs; no MoL blockchain guidance; Tribunal acquittals [11].	**Blockchain CoC** (Igonor et al., 2025): Smart-contracts hash + timestamp each custody step.	Admissible s.65B; MoL note suffices; no single-custodian trust.	14/17 jurisdictions
C4	**Cross-Border MLAT Latency.** MLAT delays 6 wk–10 mo; BD: $<$50/yr; Budapest Convention not ratified.	No US/EU CSP data-sharing; no specialist digital evidence track in MLAT.	**CLERMS** (Akilal & Kechadi, 2025): NLP pre-validates MLAT requests; catches errors pre-submission.	Low-cost MoFA/MoL change; faster preservation; fewer wasted requests.	40–70% MLAT reduction
C5	**Forensic Readiness Deficit.** No pre-incident logging; reactive posture; endpoint tools; primary logs never captured.	BTRC & CSA 2023 impose no logging standard or pre-incident config in BD.	**Self-Hosted CF** (arXiv 2510.21246, 2025): NTMC integration; no CSP cooperation; existing intercept authority.	Pre-incident capture; within legal authority; reduces investigation costs.	60% cost reduction

CIAF=Cloud Intel.&AI Forensics; CoC=Chain of Custody; MoL=Ministry of Law
CLERMS=Cross-border LE Req.Mgmt.; MoFA=Min.Foreign Affairs; NTMC=Natl.Telecom Mon.Ctr

B. Inclusion and Exclusion Criteria

A paper was included if it met four conditions: it addressed at least one of the five challenge dimensions derived from RQ1 and RQ2; it was published between 2021 and 2026; it was available in full text without a paywall; and it was in English. The paywall criterion deserves a brief note. This review is itself open-access, and including paywalled sources would undermine reproducibility—readers checking the synthesis against the cited papers should be able to read those papers. Papers were excluded if they dealt only with physical-device forensics and had nothing to say about the cloud context, or if they were conference abstracts without a methodology section. The latter exclusion is conservative but defensible: abstracts do not provide enough information to assess the quality of the underlying work.

C. PRISMA Screening Process

Of 782 initial records, 248 duplicates were removed. Screening the remaining 534 by title and abstract excluded 486 as outside scope. The 48 full-text papers were read in full: 18 were paywalled, and 14 proved too tangentially connected to any of the five clusters to justify inclusion. The final 16 studies are all open-access, span 2021–2026, and each addresses at least one challenge dimension directly. Fig. 1 shows the full flow.

D. Thematic Synthesis and Framework Construction

The 16 studies were systematically coded using the thematic synthesis procedure of Thomas and Harden [17] by a single reviewer (absence of inter-rater verification is a stated limitation). Each study was coded on three dimensions: (a) which challenge cluster(s) it addresses; (b) whether Bangladesh relevance is direct, partial, or absent; and (c) whether it proposes an AI or automation-based resolution tool. The comparative matrix is in Table I. Each cluster was then matched to the highest-evidenced resolution tool (RQ3) based on empirical metrics (F1-scores, jurisdiction counts, latency and cost reductions), yielding the AI-CFRF-BD framework (RQ4). Table II presents the complete taxonomy.

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